



MINERALS

Some chemical compounds exist as gases such as carbon dioxide, or liquids such as water. Others form solids known as minerals. A mineral is composed of one or more elements in fixed proportions, usually with a distinctive crystal structure. There are more than 4,500 minerals, including silica, a compound of two elements—silicon and oxygen. In its pure form, silica occurs as quartz. Flint, used to make this ancient arrowhead, is made up of fine-grained quartz.



METALS

The shiny elements known as metals are made of identical atoms and are good conductors of heat and electricity. Most metals react with other elements to form minerals known as metal ores. For example, iron combines with oxygen to form iron oxide, the main ingredient of iron ore—but more familiar to us as rust.

The most abundant element in the Universe is hydrogen.



NATURAL PAINT BOX

Many metal-bearing minerals are colorful, such as the green copper compound malachite, and the red mercury-sulfur compound cinnabar. In the past, they were ground up and used as pigments in paints. Some, such as blue lapis lazuli, were rare and highly valued, and used to adorn religious paintings and manuscripts.



MINERALS AND ROCKS

The planet is made largely of rocks, which are all created from combinations of minerals, which in turn are made from elements. This granite is made of the minerals quartz, feldspar, and mica. These contain the metallic elements potassium, sodium, aluminum, iron, and magnesium, plus silicon and oxygen.